HIGH PERFORMANCE COMPUTING FOR SCIENCE AND ENGINEERING

Harvard University
CS 2050

Lecture 20

April 9, 2026

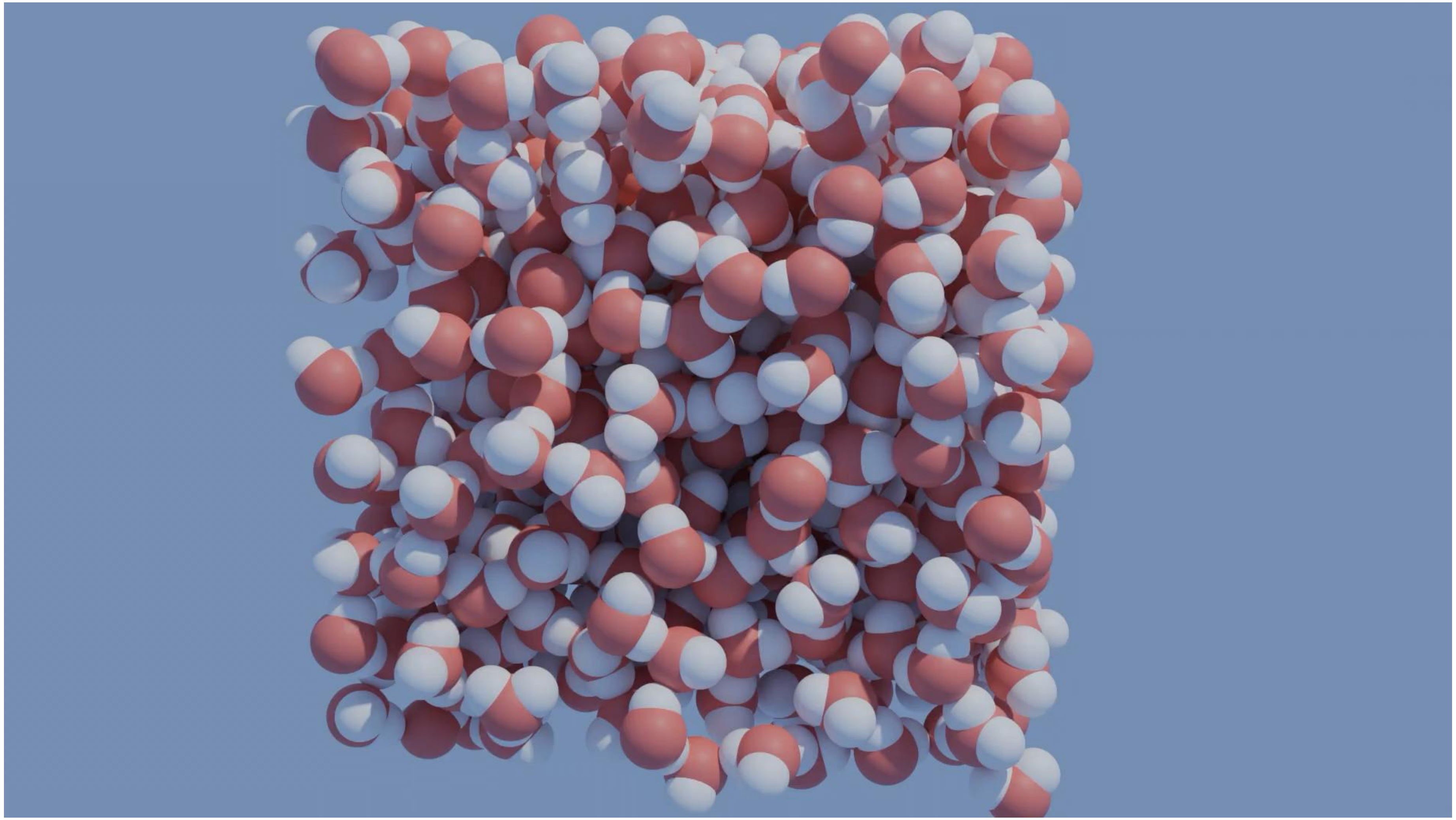
PREPARE FOR EXAMPLES

- Please pull the latest `lecture-examples`

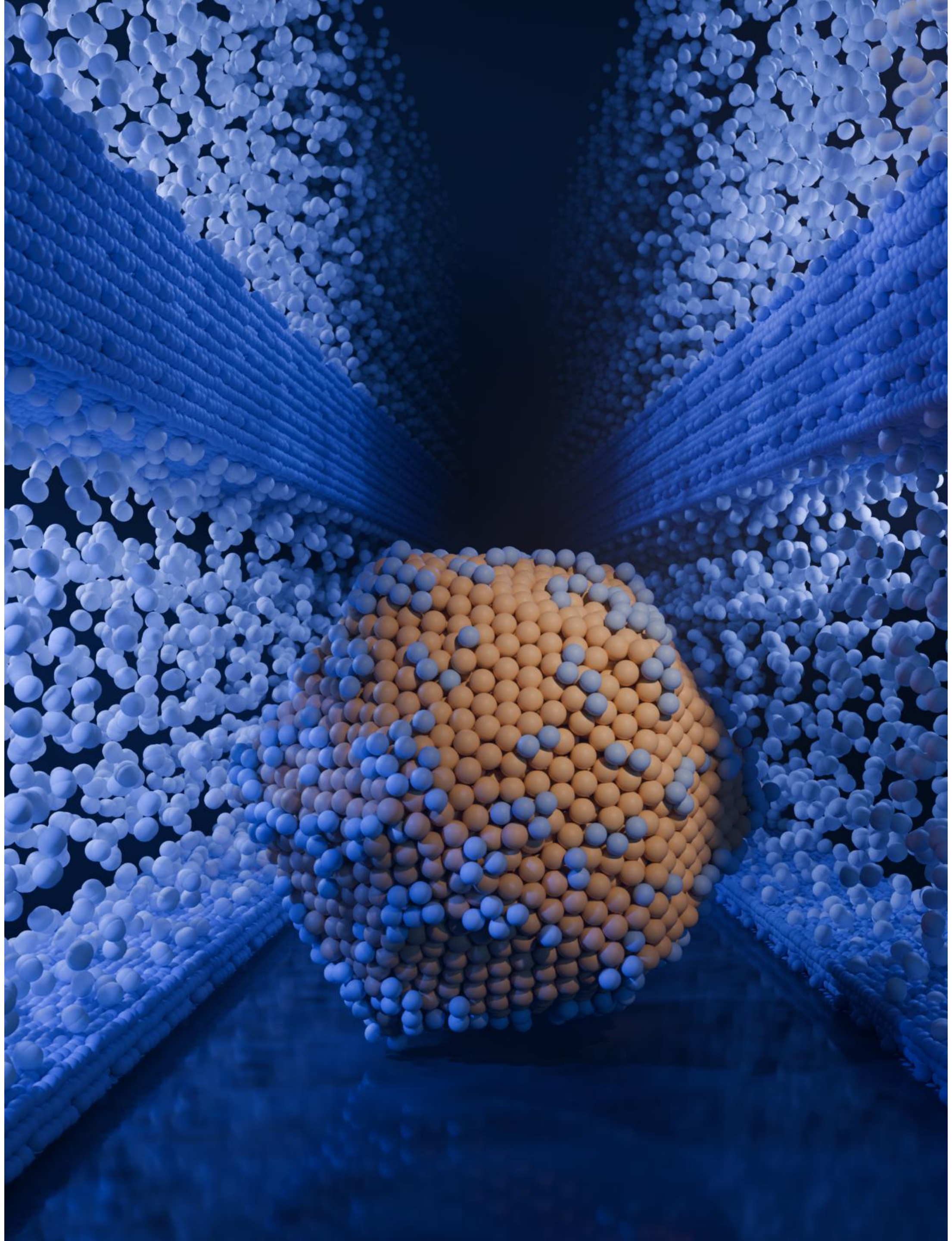
MOLECULAR DYNAMICS SIMULATION

Research Examples

FROM MOLECULES

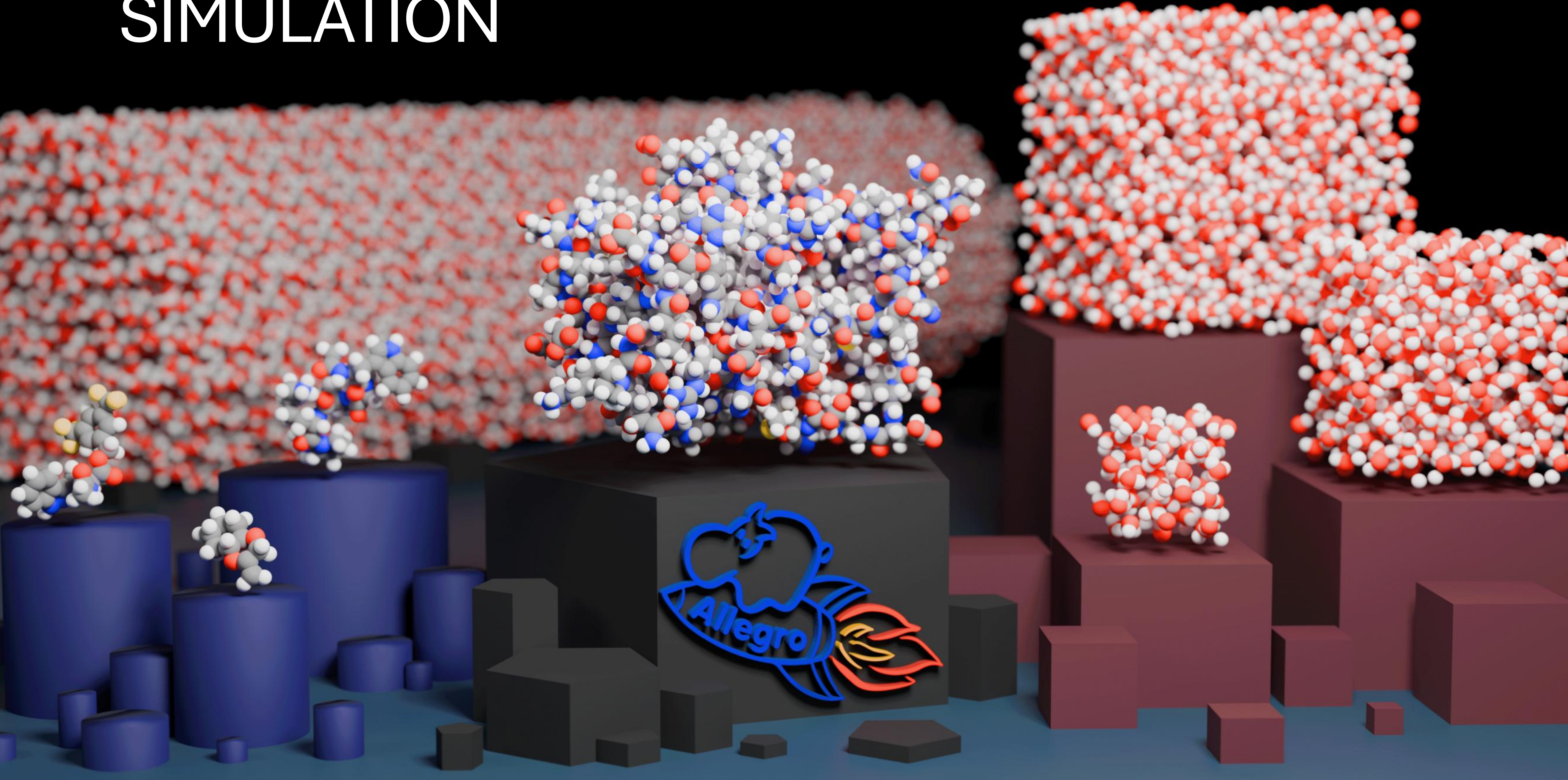


TO MATERIALS



Nanoparticle structure
courtesy of Cameron J. Owen

LARGE SCALE AND QUANTUM ACCURATE SIMULATION

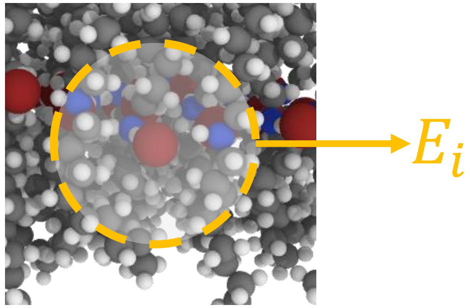


MACHINE LEARNED FORCE FIELDS

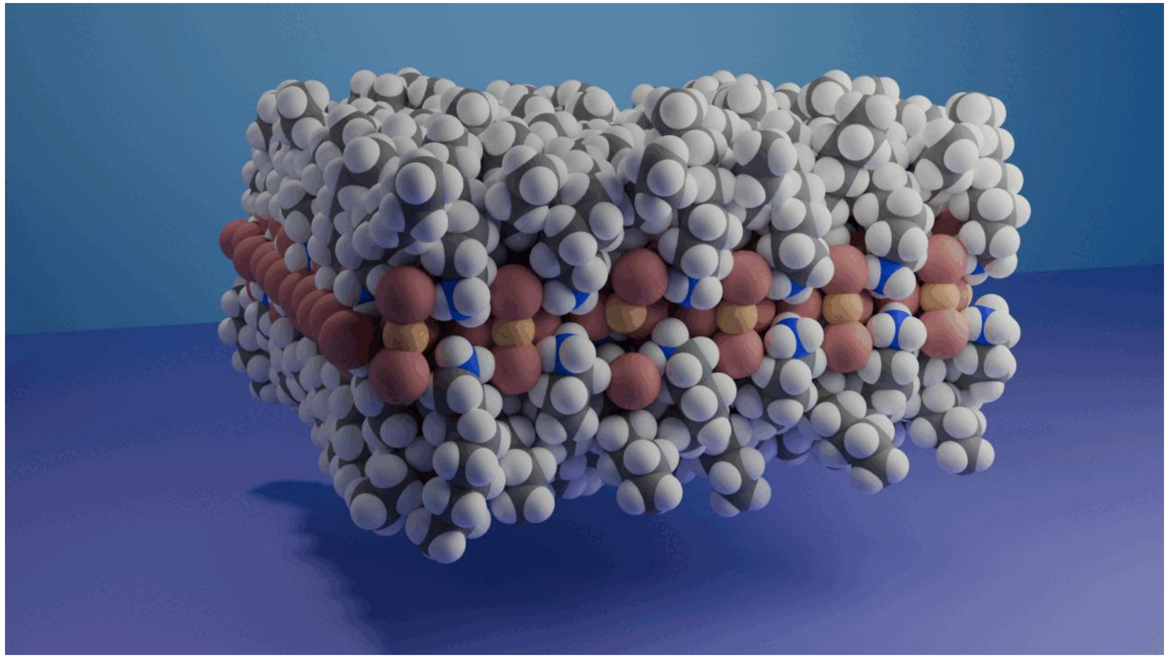
$$E_{\text{total}} = \sum_i E_i$$

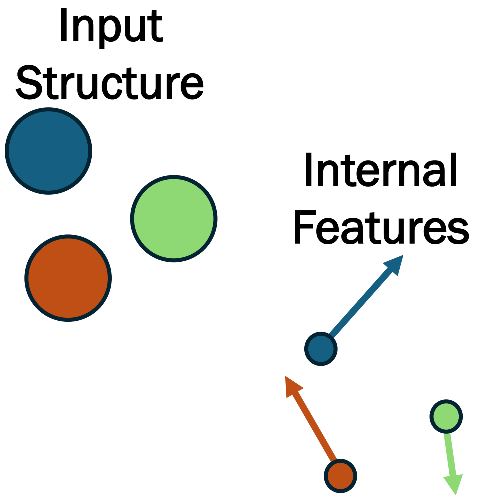
$$F = -\nabla E_{\text{total}}$$

Conservative forces



Deep Equivariant Neural Network Force Fields

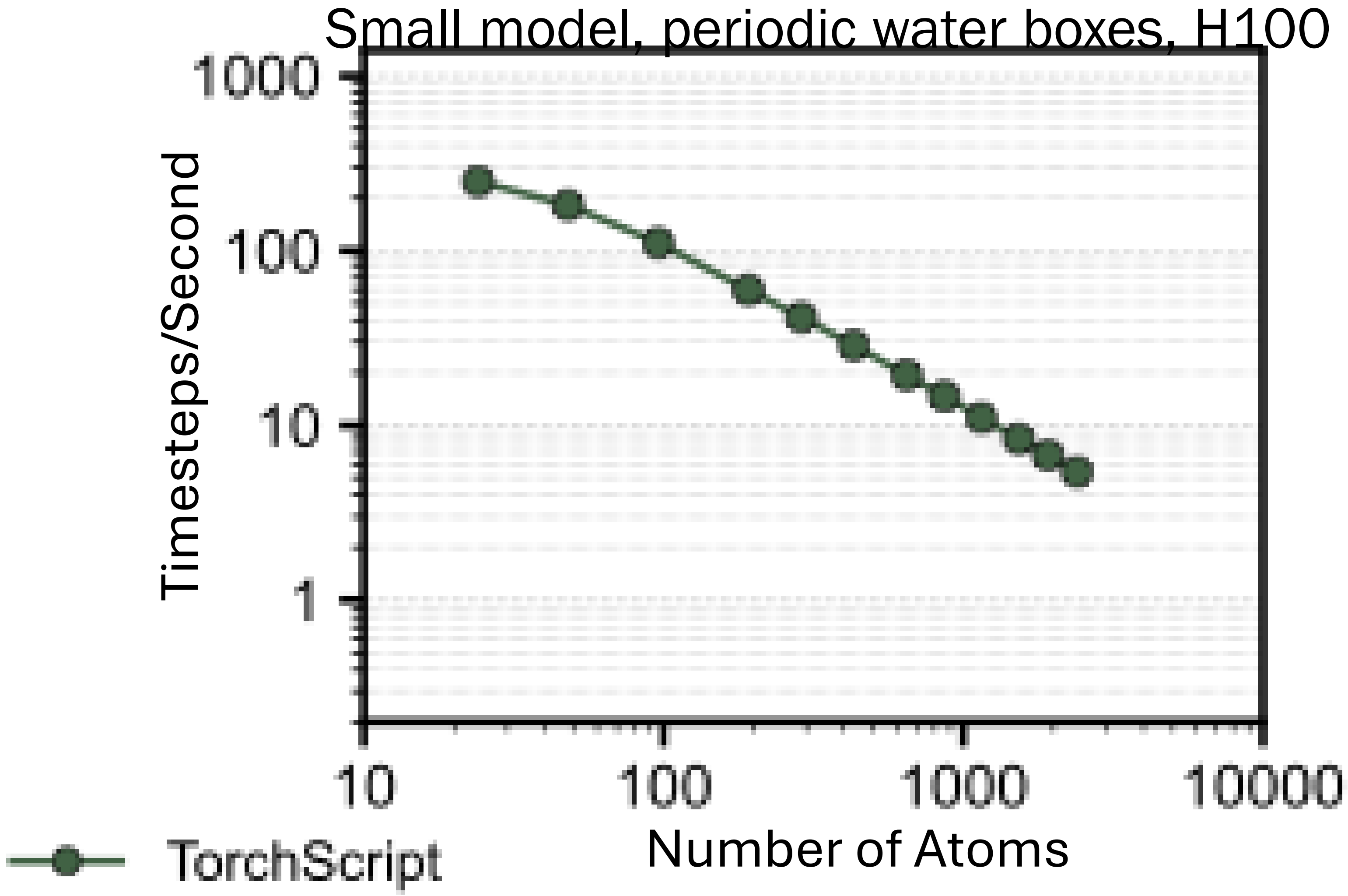




The internal features of the model are E(3)-equivariant, enforced via architecture and the use of the Clebsch-Gordon Tensor Product

Materials Intelligence Research

REDUCING OVERHEAD AND IMPROVING EFFICIENCY



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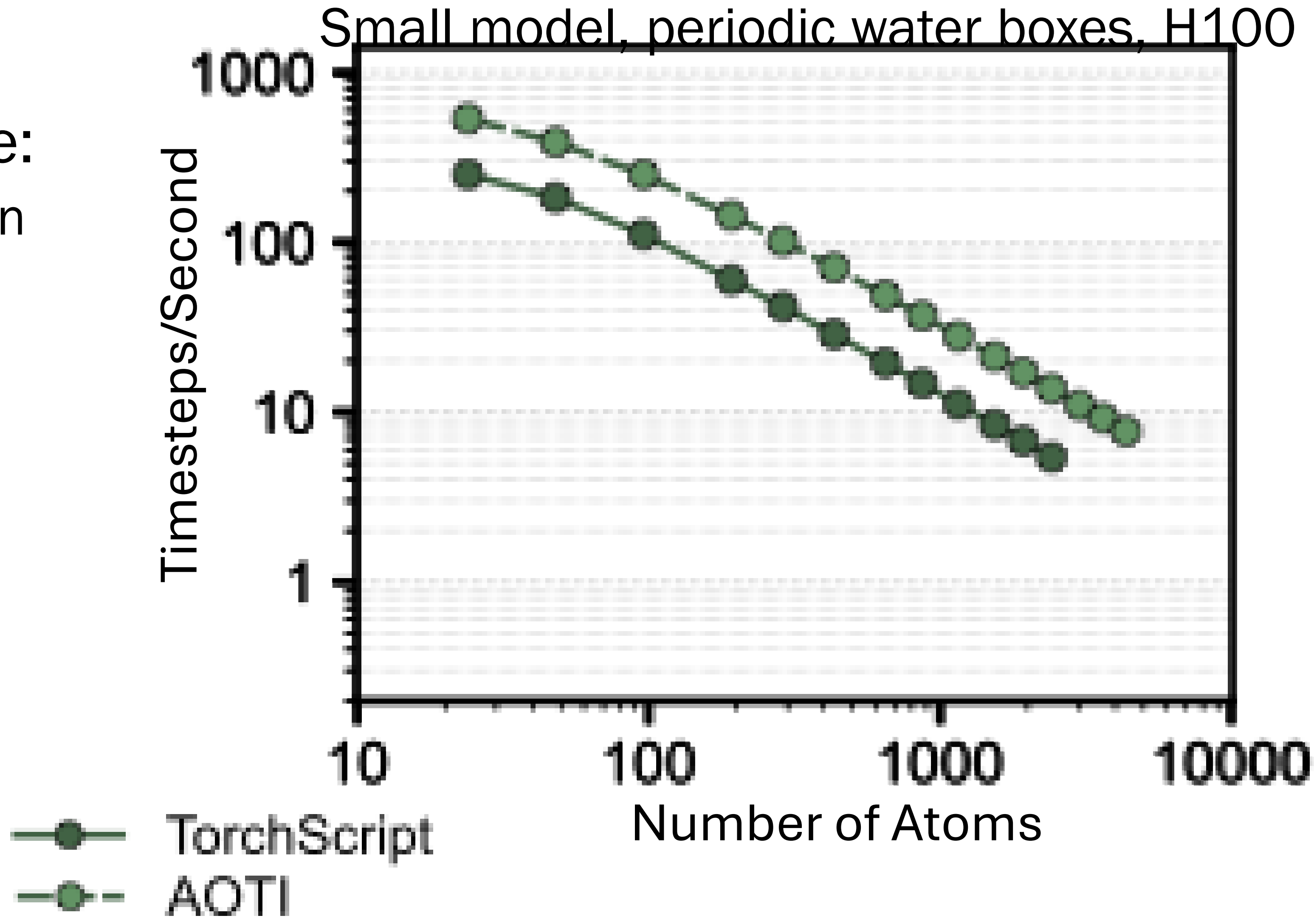
REDUCING OVERHEAD AND IMPROVING EFFICIENCY

- With Torch's new export and AOT-Inductor functionality we:
  - Perform full-graph compilation
  - Use dynamic shapes
  - Tune kernel settings for the available hardware



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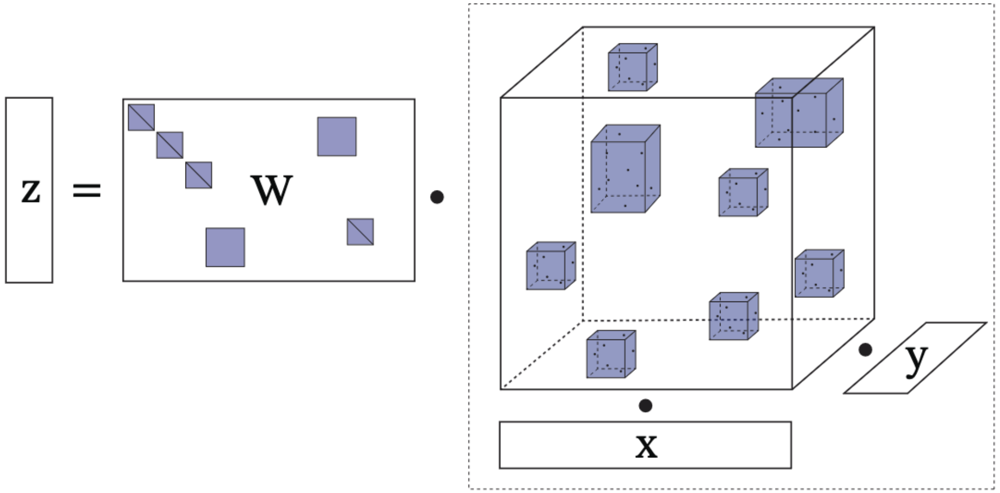
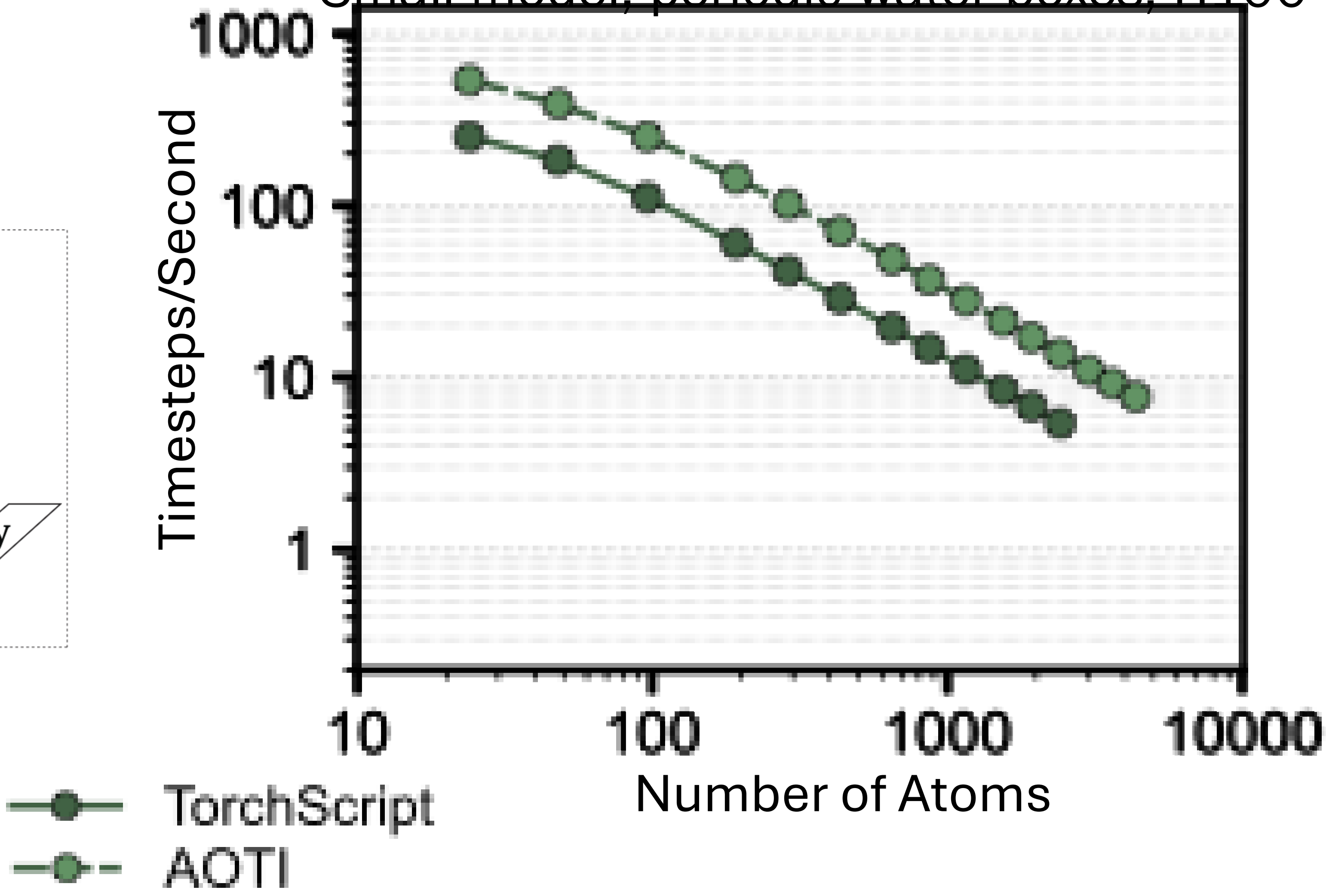
REDUCING OVERHEAD AND IMPROVING EFFICIENCY

The Clebsch-Gordon Tensor Product



Small model, periodic water boxes, H100



Bharadwaj, V. et al. (2025). An efficient sparse kernel generator for o (3)-equivariant deep networks. In *2025 Proceedings of the Conference on Applied and Computational Discrete Algorithms (ACDA)* (pp. 32-46). SIAM

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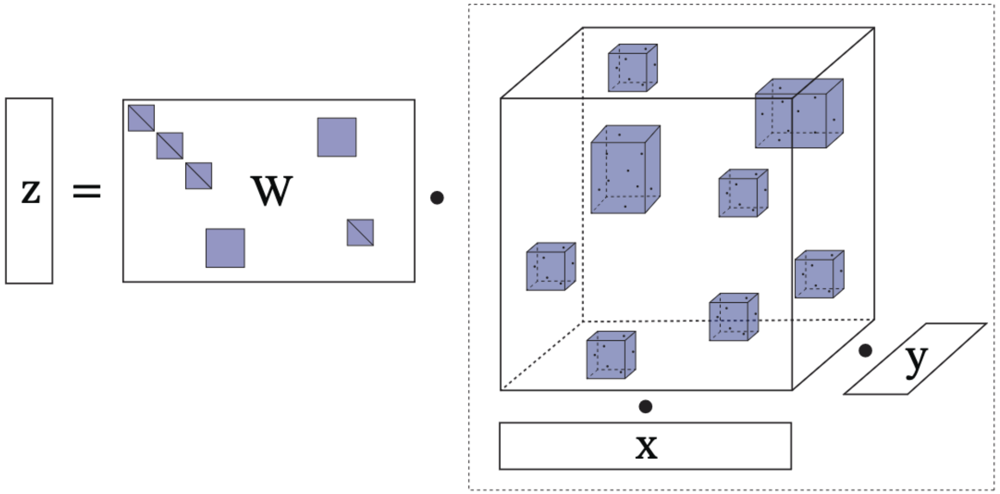
REDUCING OVERHEAD AND IMPROVING EFFICIENCY

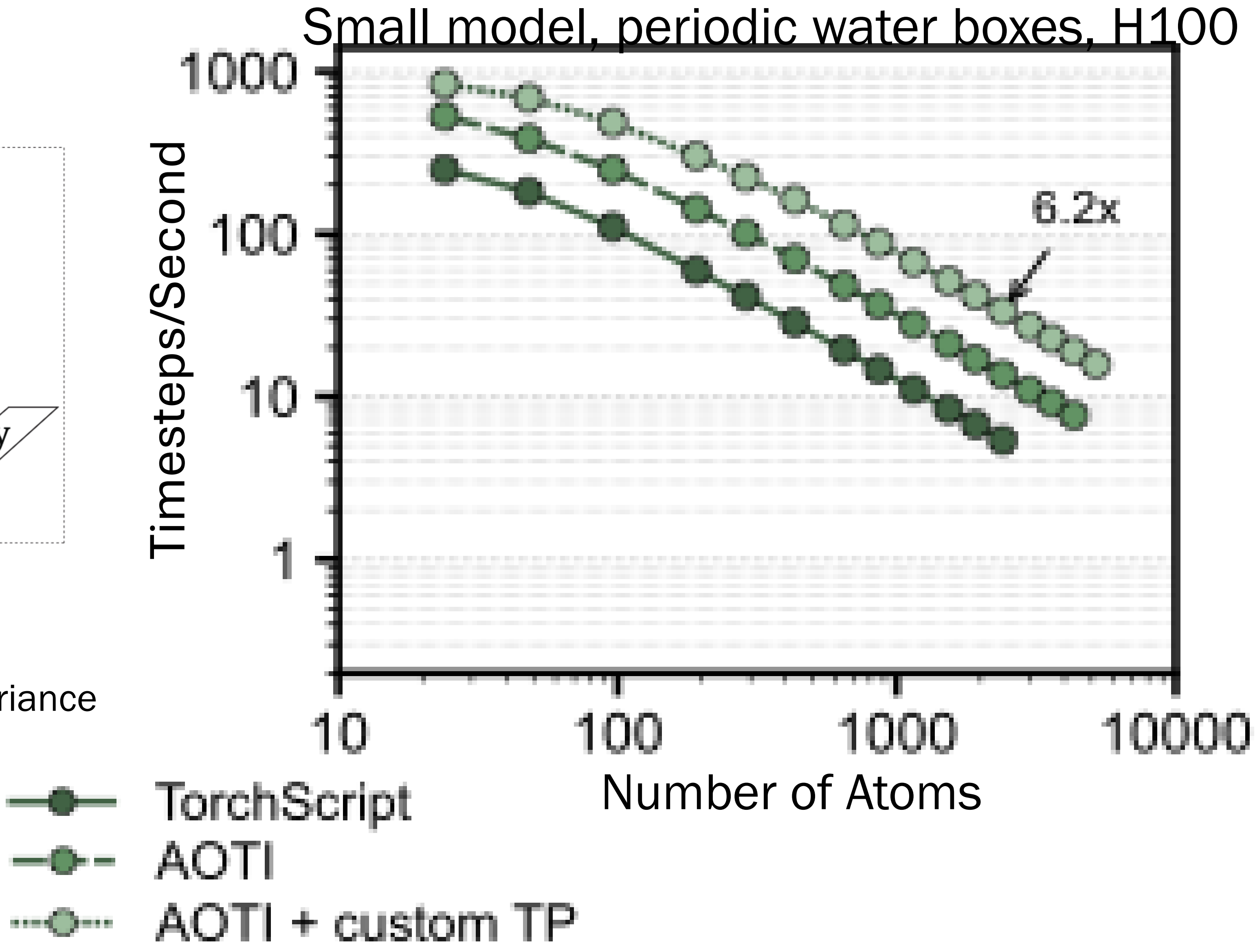
The Clebsch-Gordon Tensor Product



Replace with custom tensor-product kernels:

NequIP: OpenEquivariance

Allegro: Custom Triton, NVIDIA cuEquivariance



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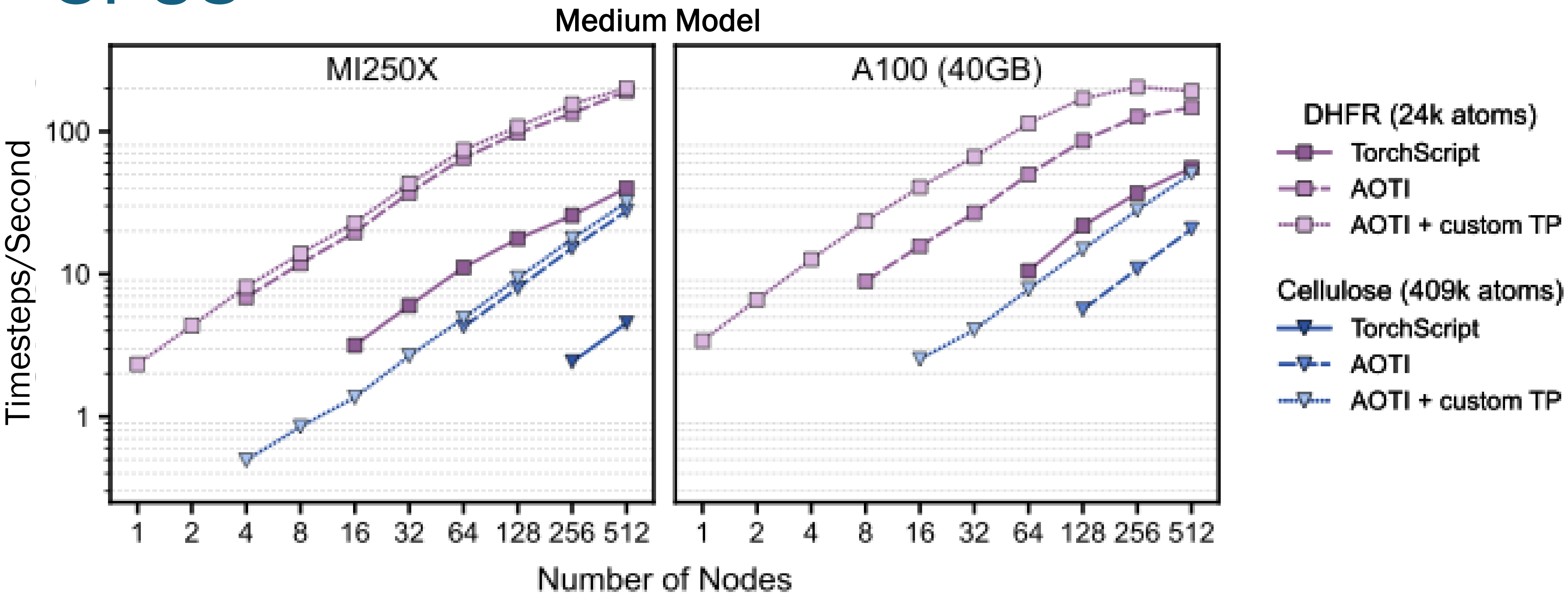
ALLEGRO IN LAMMPS SCALES TO 1000S OF GPUS



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THE EMERGING POTENTIAL FOR SOLID-STATE COOLING

Barocaloric materials couple pressure with temperature

This property enables them to drive a heat cycle in the **solid-state** by leveraging order-disorder phase transitions.

In general, we would like to optimize the transition's:

- Reversibility
- Entropic change
- Pressure sensitivity
- Hysteresis

to improve the efficiency and efficacy of the heat cycle

I computationally examine two barocaloric materials to understand their entropic behavior

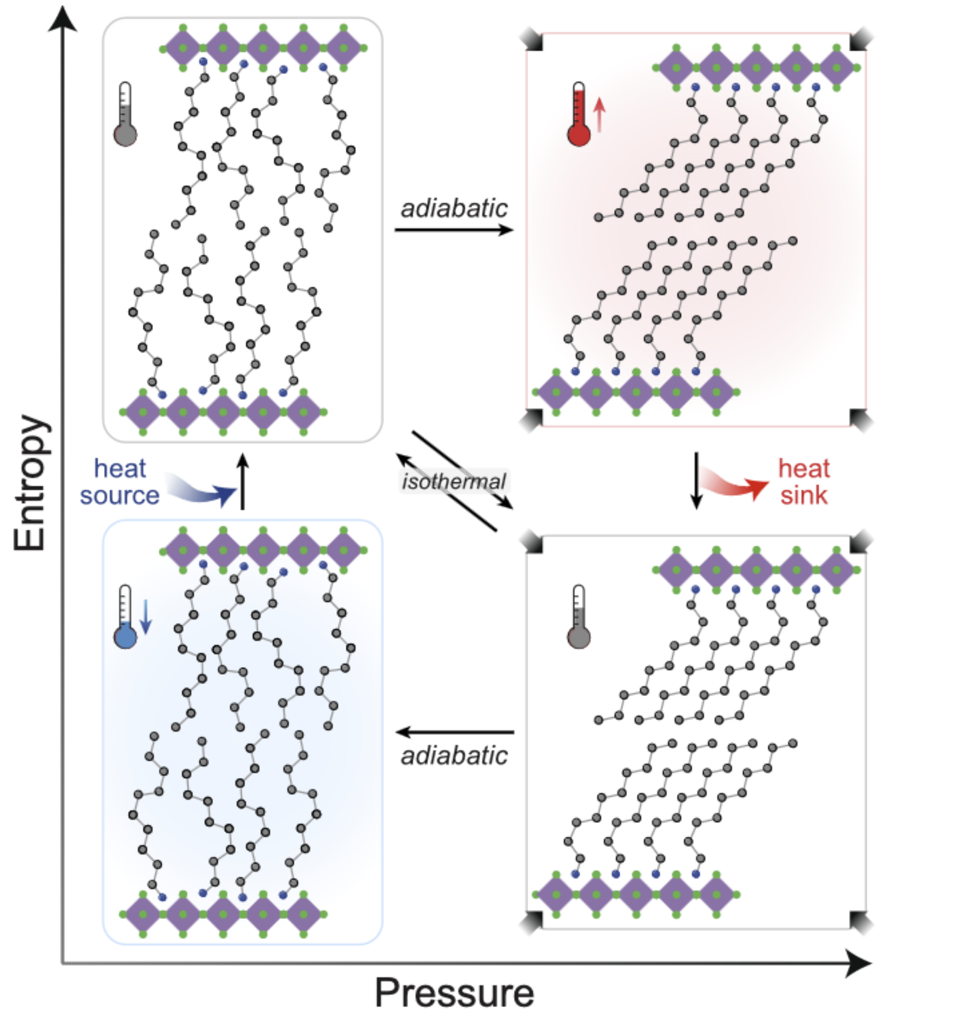

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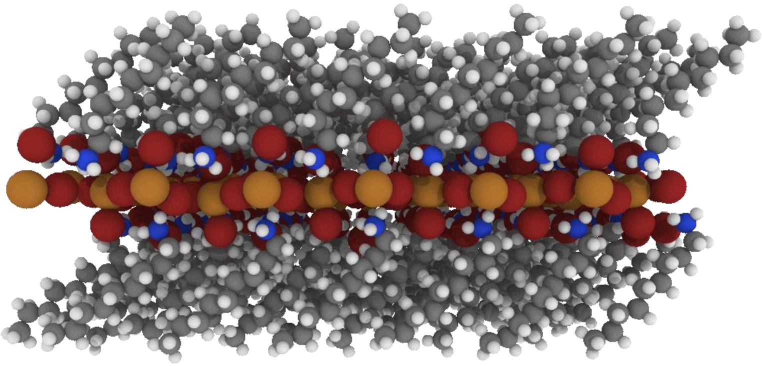
Figure adapted from Seo, J., Mason, J. A., *et al.* (2022). *Nat. Commun.*, *13*(1), 25

WHAT DRIVES THE DIFFERENCE IN $\Delta S_{\mathrm{TR,C9}}$

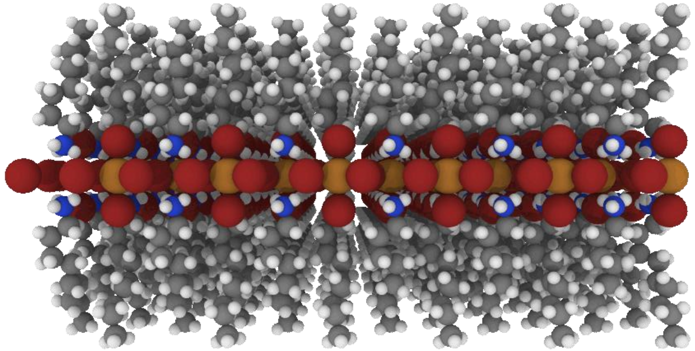
$(C_9)_2CuBr_4$



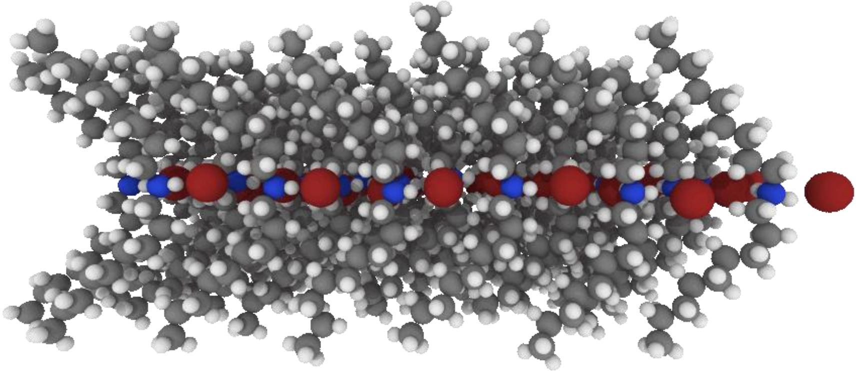
Heating

Cooling

$\Delta S_{\mathrm{tr,C9}}$
$= 25.5\ \mathrm{J\ mol^{-1}\ K^{-1}}$



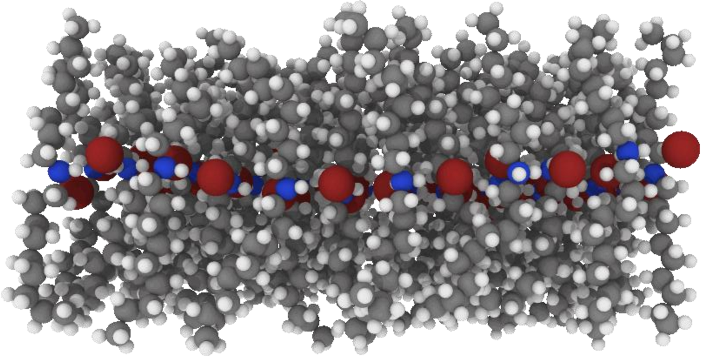
dC_9-Br



Heating

Cooling

$\Delta S_{\mathrm{tr,C9}}$
$= 44.4\ \mathrm{J\ mol^{-1}\ K^{-1}}$



C
N
Br
Cu
H

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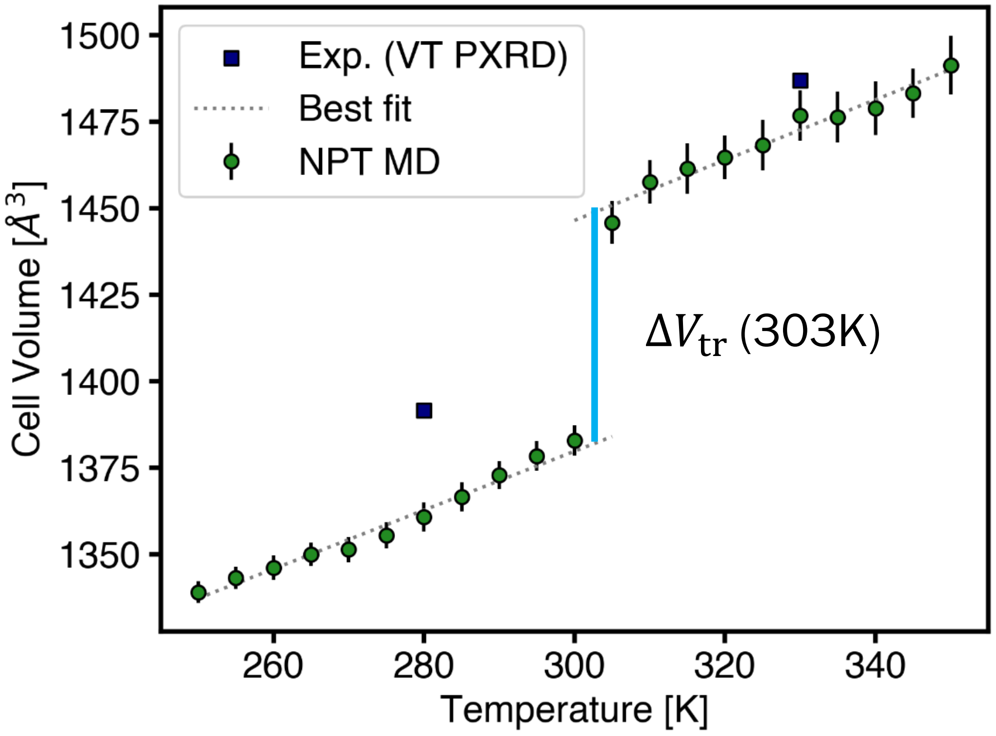
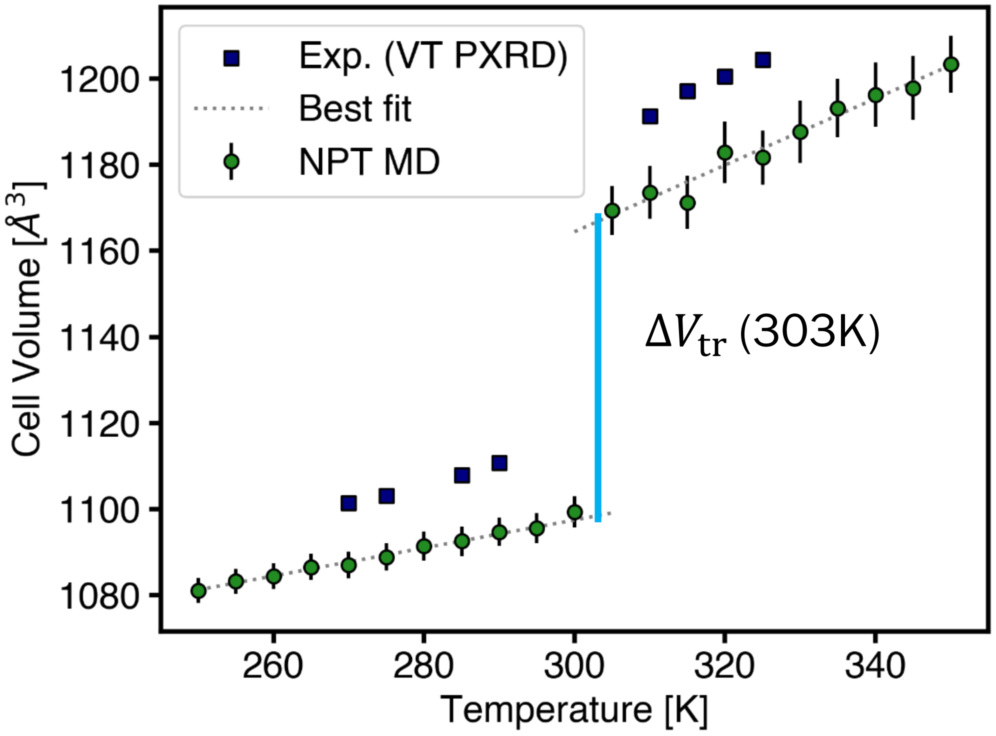
NPT MD SIMULATIONS CAPTURE THE TEMPERATURE-VOLUME RELATIONSHIP

$(C_9)_2CuBr_4$



Best fits estimate the $\Delta V_{\mathrm{tr}} = 4.61\%$
Experiment in [1] is $\Delta V_{\mathrm{tr}} = 4.0\%$

dC_9-Br



Best fits estimate the $\Delta V_{\mathrm{tr}} = 6.23\%$
Experiment best fit is $\Delta V_{\mathrm{tr}} = 6.2\%$

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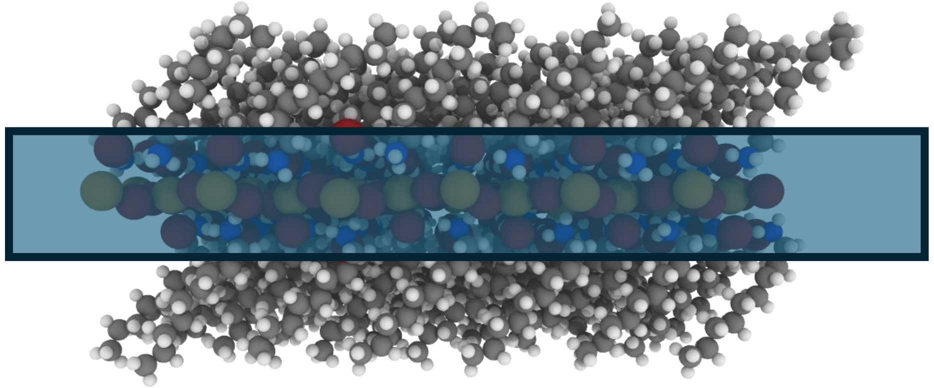
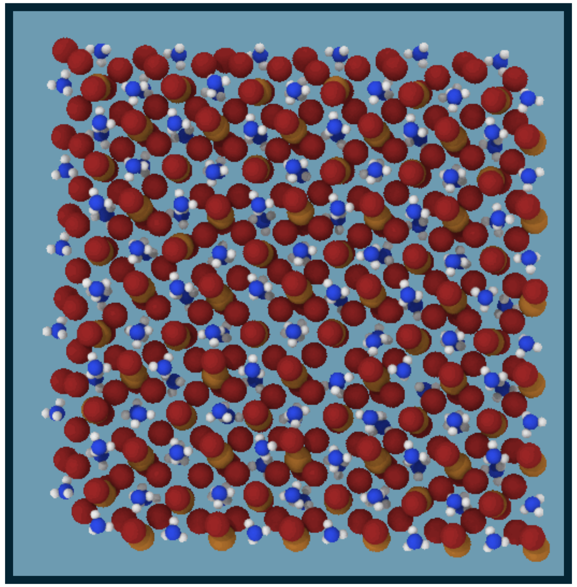
[1] Seo, J., Mason, J. A., *et al.* (2022). *Nat. Commun.*, *13*(1), 2536.

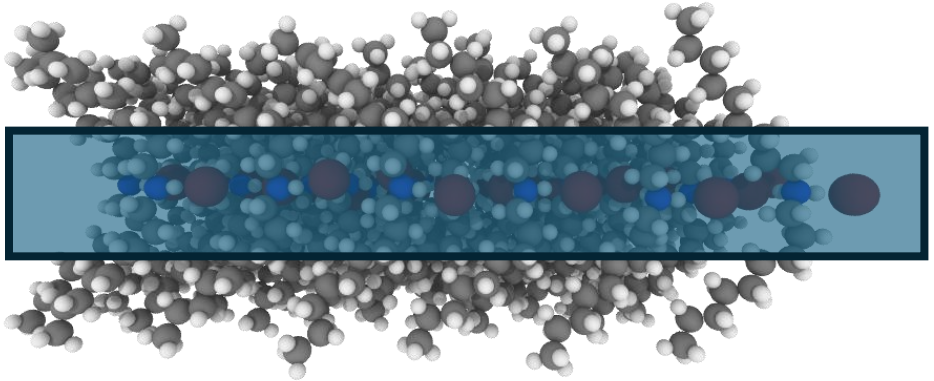
INVESTIGATING THE CHAIN ANCHORS AND INORGANIC LAYER

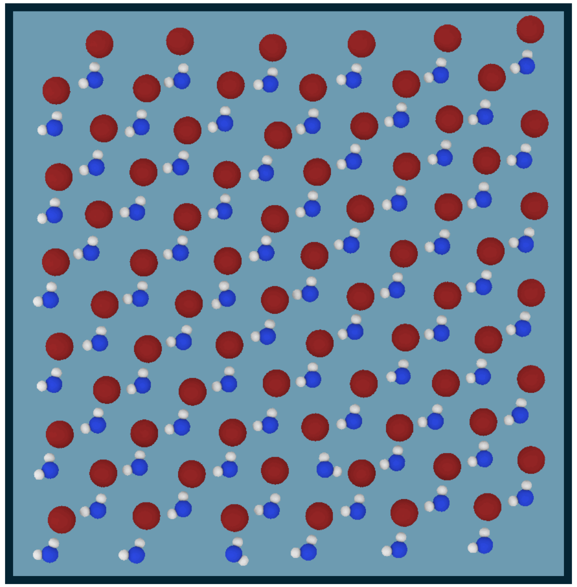
$(C_9)_2CuBr_4$





dC_9-Br





C
N
Br
Cu
H

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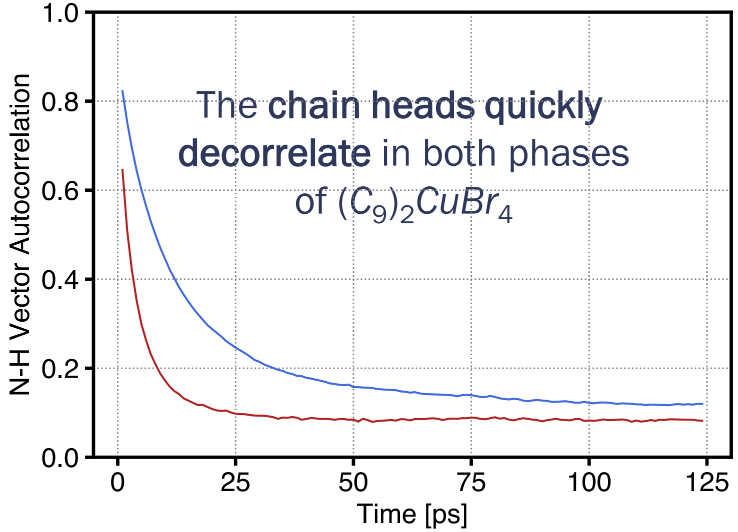
A STARK DIFFERENCE IS THE STABILITY OF H-BONDING

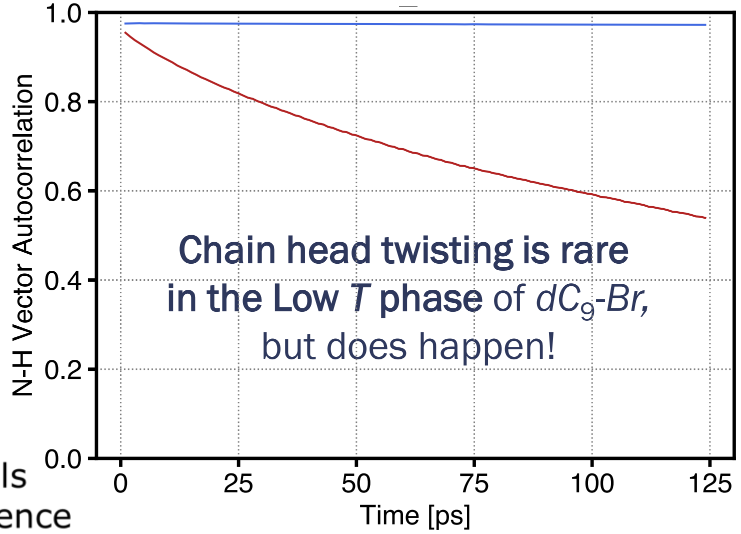
Autocorrelation of N-H bond vectors

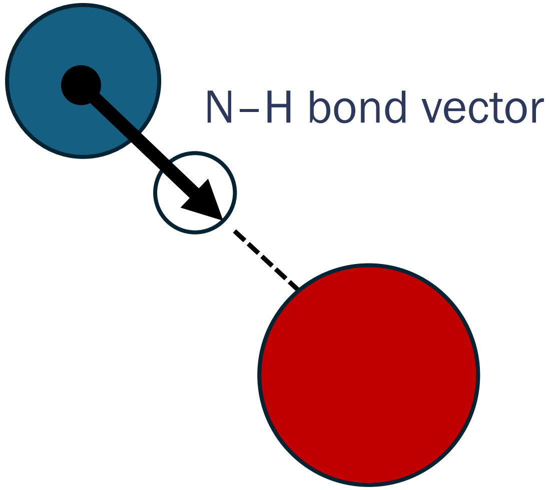
T = 280K *T* = 350K

$(C_9)_2CuBr_4$

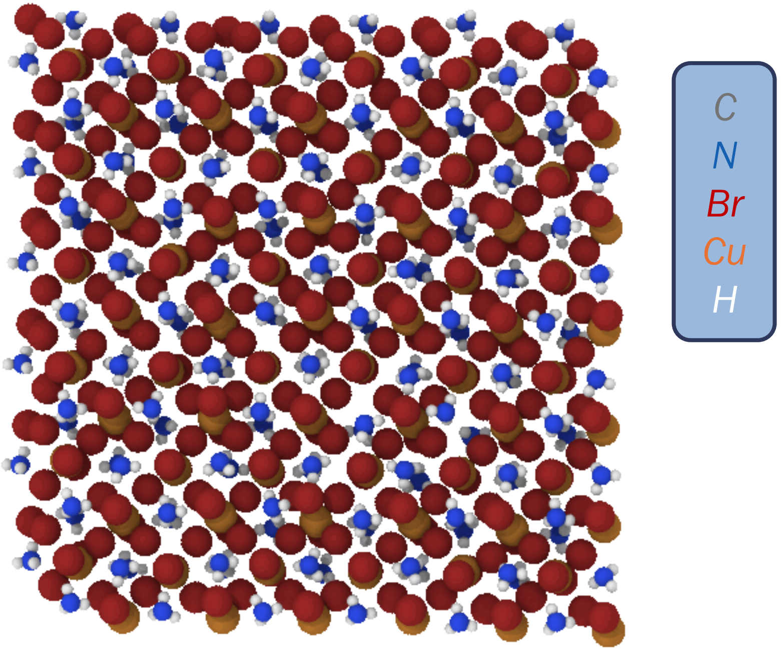


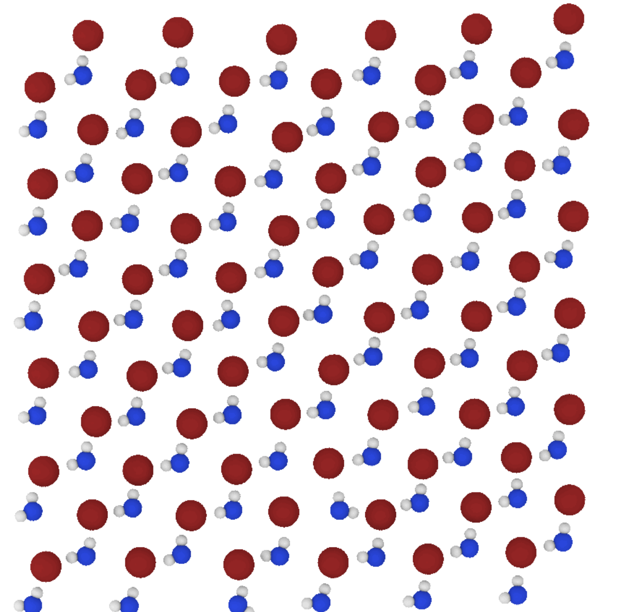
dC_9-Br





The chain anchor forms H-bonds with Br in the organic layer





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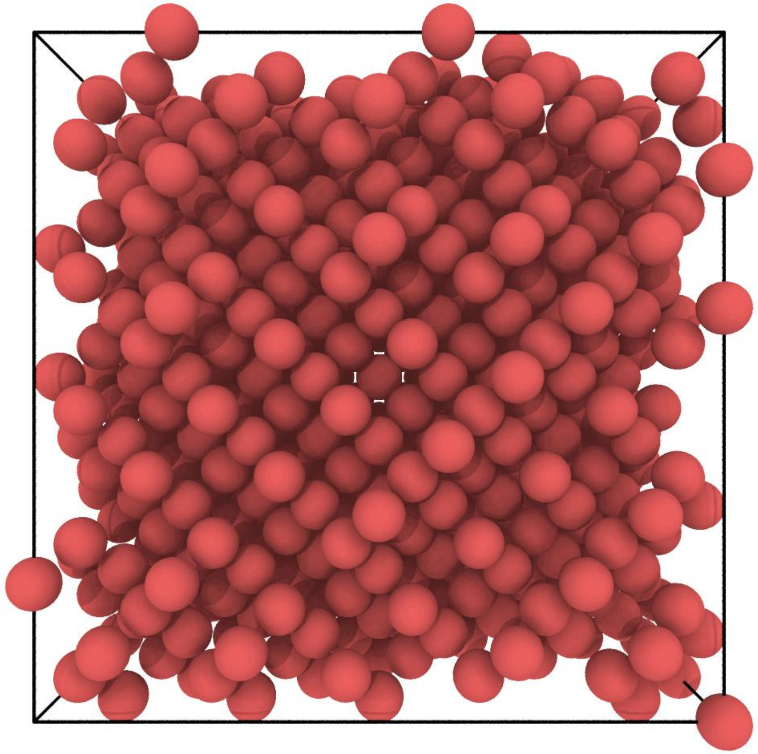
MOLECULAR DYNAMICS SIMULATION

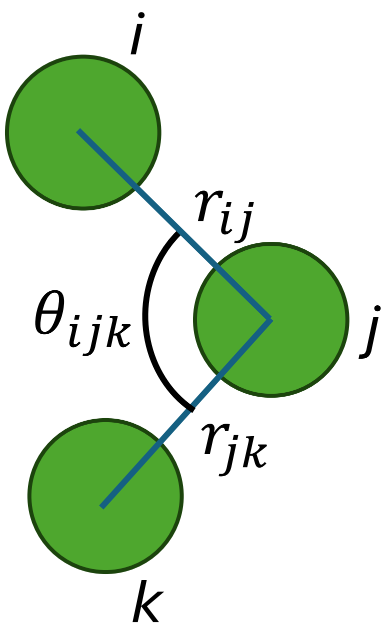
Parallelization Techniques



SHARED MEMORY PARALLELISM

- Many operations in the MD Engine are loops over each atom:
  - State updates (position, velocity, etc.)
  - Force calculations over pairs, triplets of atoms
- We just need to handle race conditions:

$$F_j = \phi_2(r_{ij}) + \phi_2(r_{ji}) + \phi_3(r_i, r_j, r_k) + \cdots$$



POINTWISE UPDATES

- Integration Loop:

  For i in (all atoms):

  Update position array values

  Update velocity array values

- Thermostat Loop:

  For i in (all atoms):

  Apply velocity scaling factor

ENERGY/FORCE REDUCTIONS

- Energy Loop:

```
For i in (all atoms):
    For j in (neighbor atoms):
        Compute pair, triplet energy
        E_i += energy
```

- Force Loop:

```
For i in (all atoms):
    For j in (neighbor atoms):
        Compute pair, triplet force
        F_i += force
```

INCORPORATING OPENMP PARALLELISM

- Exercise 1

- The water molecules are starting to wake up.

HOW FAR DOES THIS GET US?

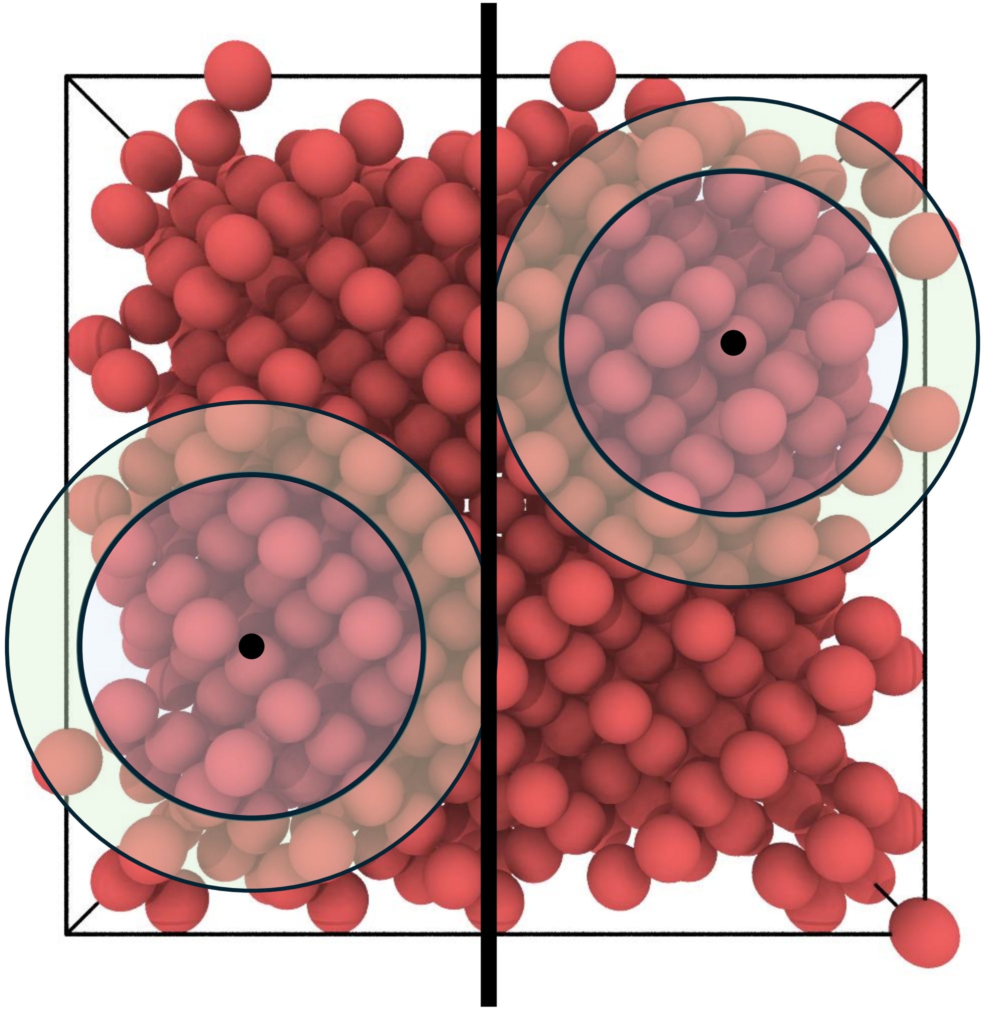
- Exercise 2

- One parallelism choice seems better for our force OpenMP reduction.

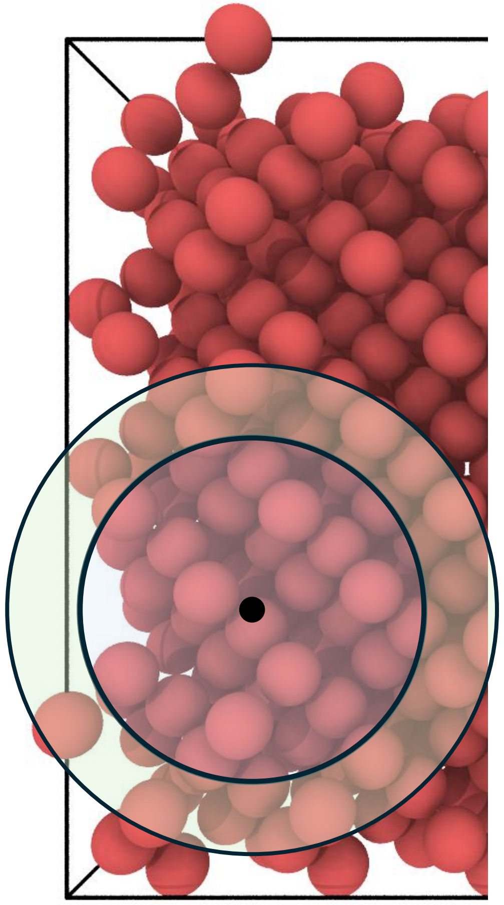
DISTRIBUTED MEMORY PARALLELISM

- We need larger scale parallelism than one node can provide
- But it is hard to split an interacting system without some communication
  - We cannot afford a lot of communication for every timestep
- We need to use some Algorithms and Physics intuition!

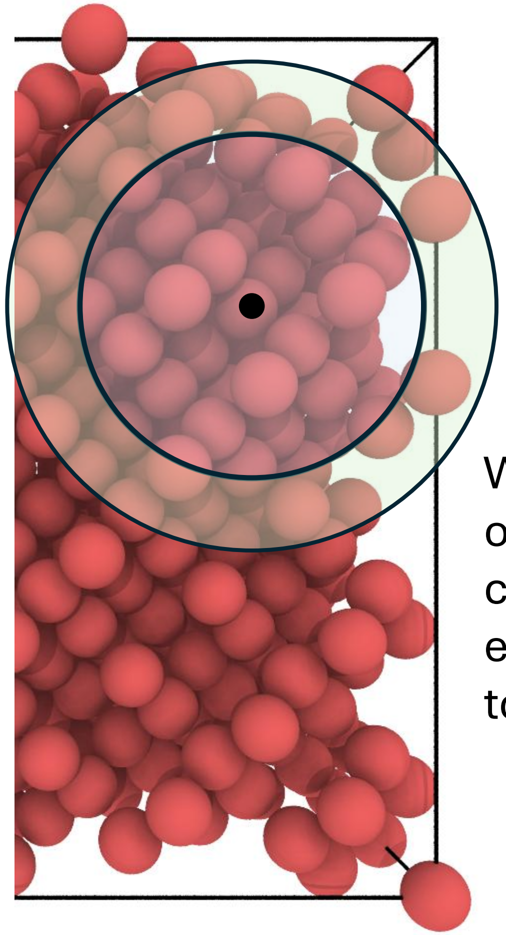
DIVIDING OUR SYSTEM



DIVIDING OUR SYSTEM



MPI Rank 1



MPI Rank 2

We use the spatial locality of the most computationally expensive calculations to minimize communication

COMMUNICATION ALGORITHMS

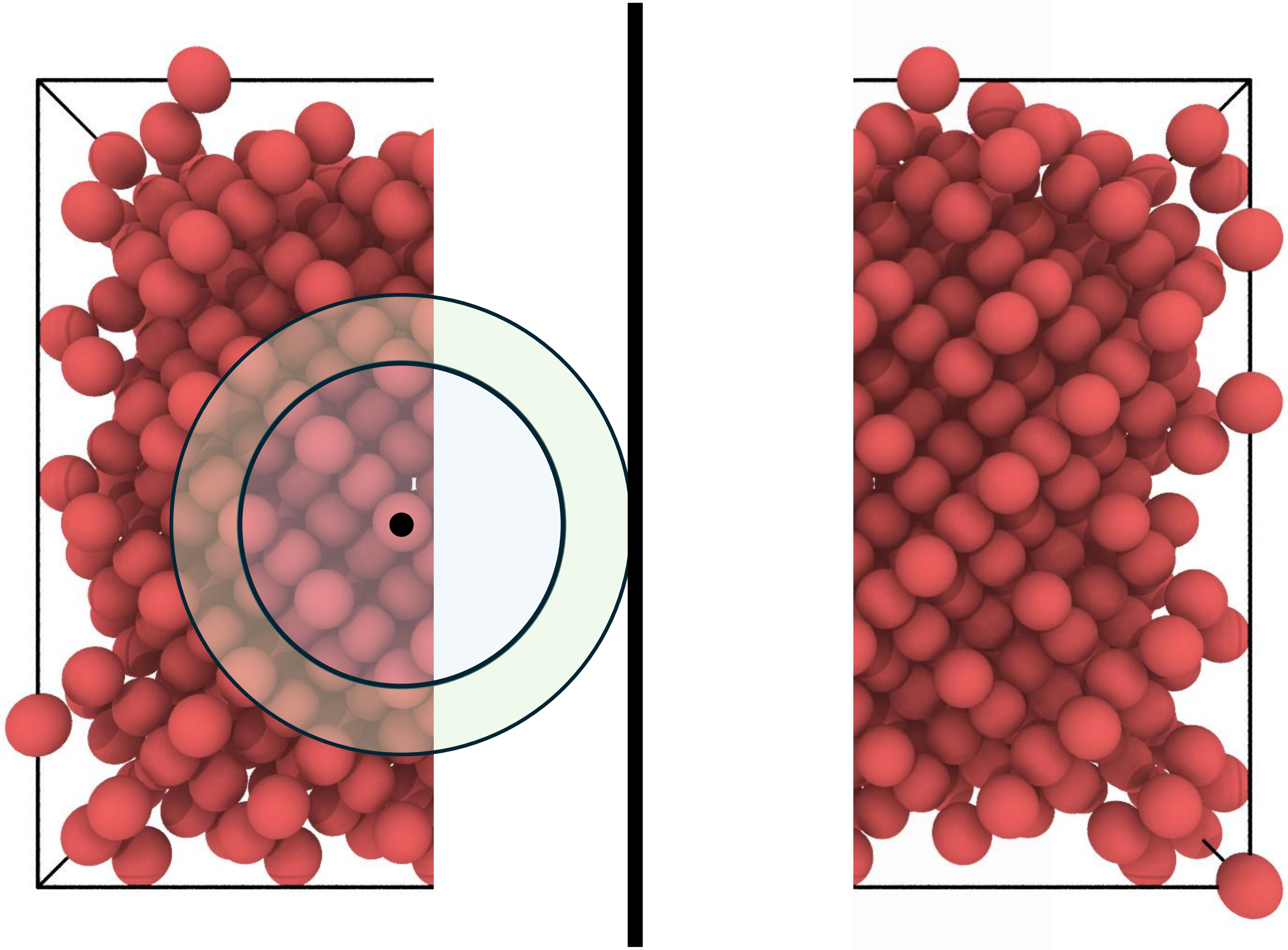
- Simpler operations:
  - Temperature is a sum over atoms → MPI AllReduce

$$T = \frac{1}{3Nk_B}\sum_{i=1}^{N} m_i v_i^2 \qquad E_{\text{potential}} = \sum_{i<j}\phi_2(r_{ij}) + \sum_{i<j<k}\phi_3(\mathbf{r}_i, \mathbf{r}_j, \mathbf{r}_k)$$

  - Logging our system has rank 0 collect all atoms → MPI GatherV

- More complex operations require more complex communication patterns

FORCES ACROSS THE INTERFACE



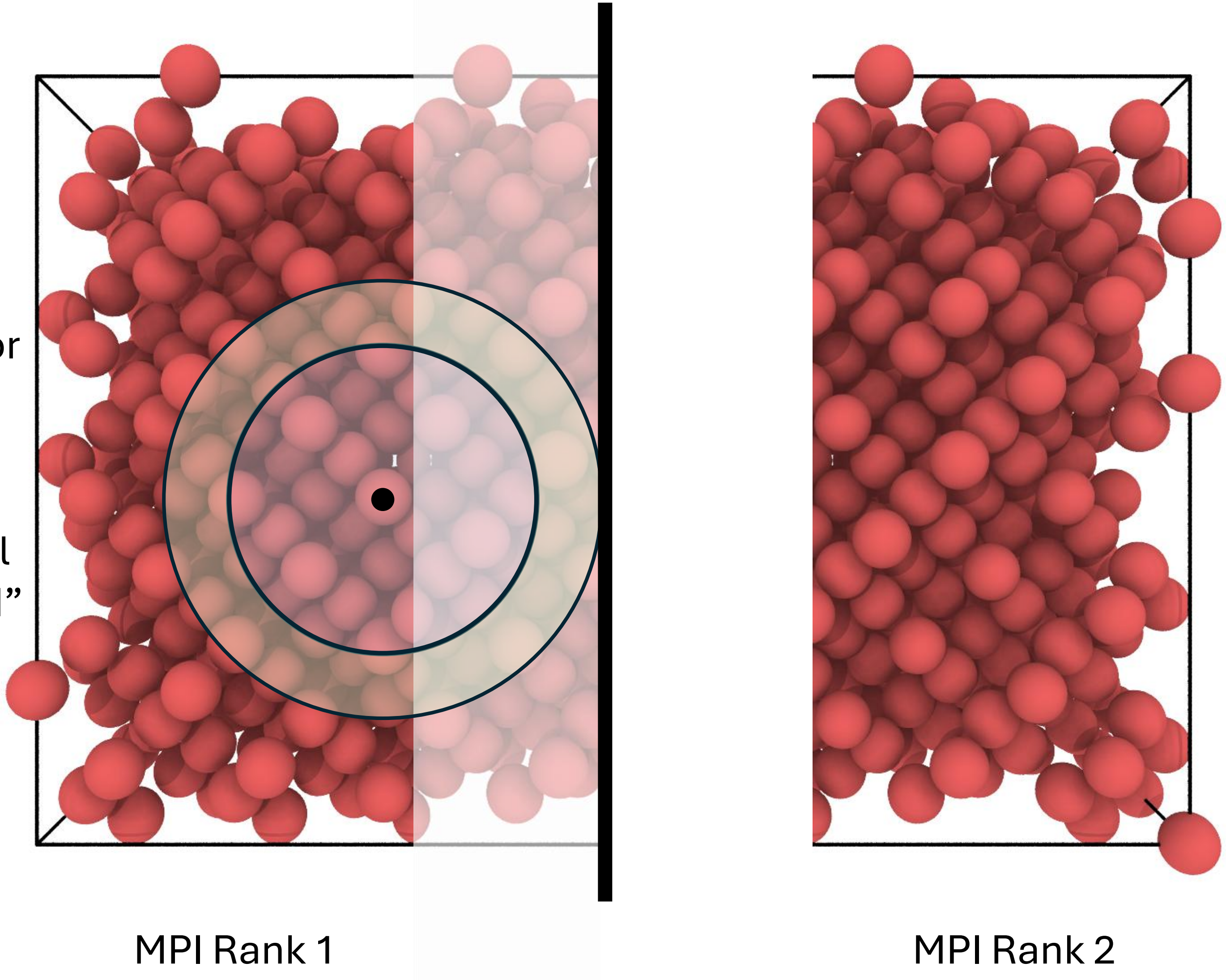
MPI Rank 1

MPI Rank 2

FORCES ACROSS THE INTERFACE

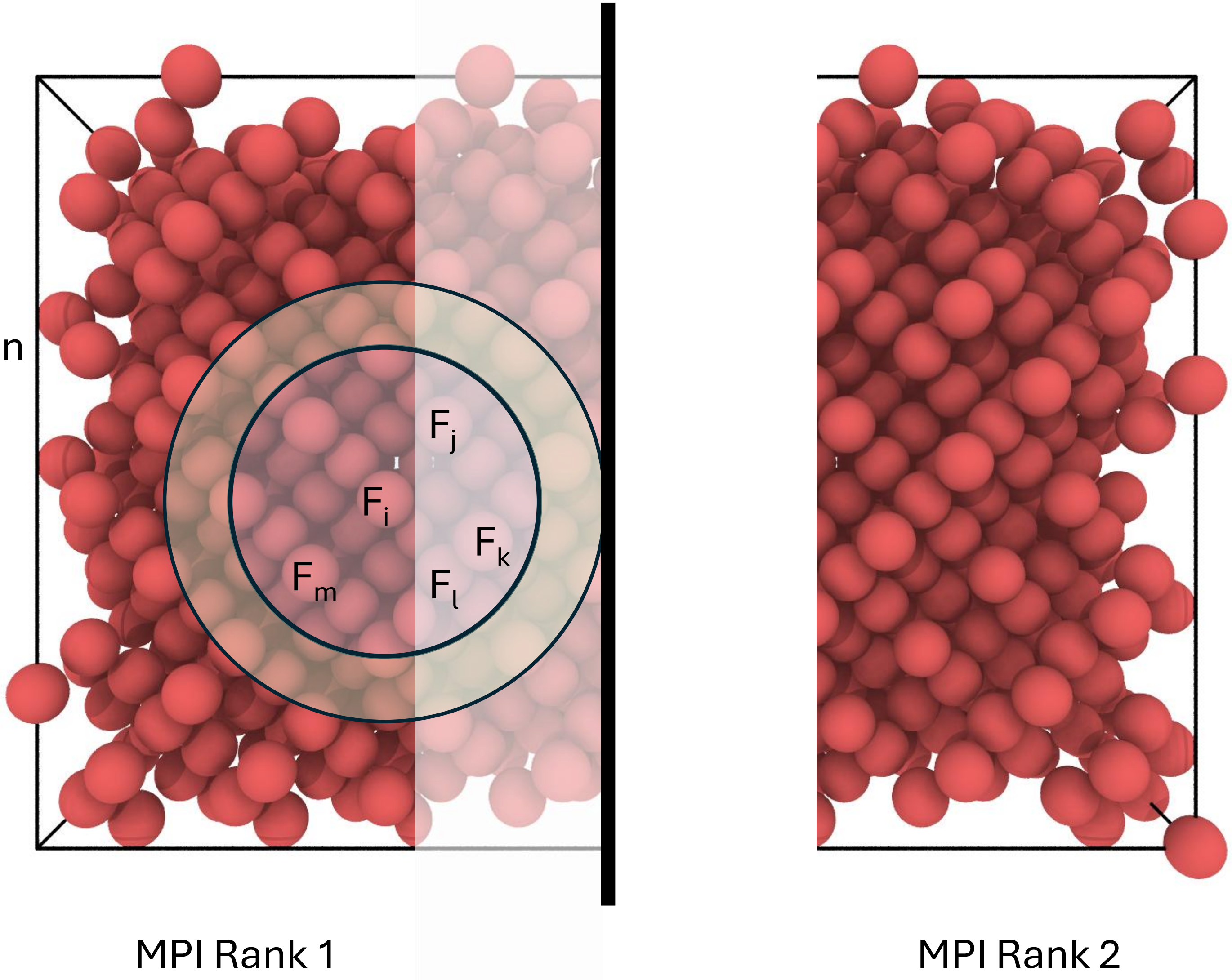
Each rank copies relevant information from its neighbor processors

We call these copies “ghost atoms”, while original atoms are ”local” or “owned”



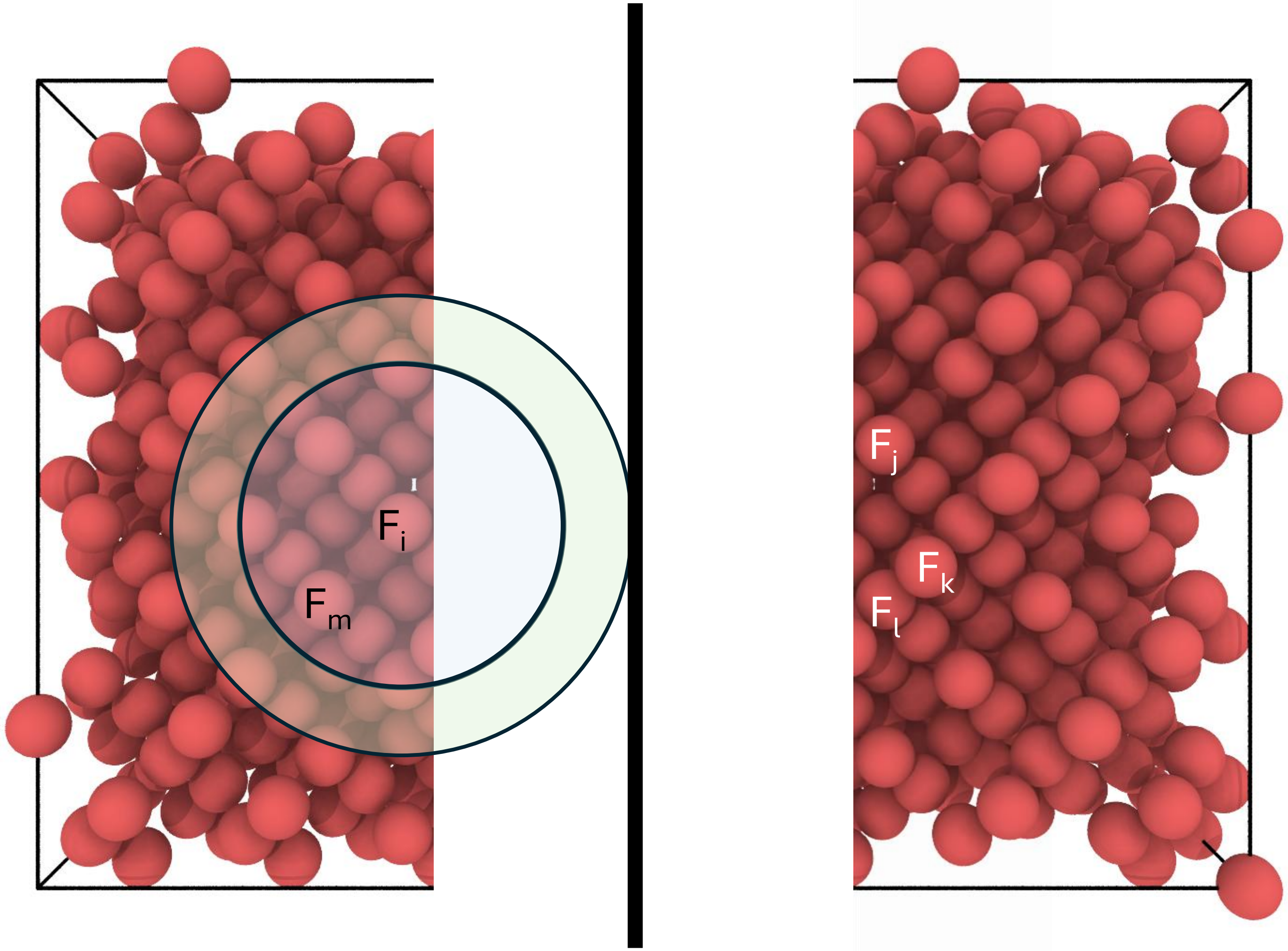
FORCES ACROSS THE INTERFACE

We compute energy, force using ghost atom information



FORCES ACROSS THE INTERFACE

We send the forces applied to ghost atoms back to their home rank



MPI Rank 1

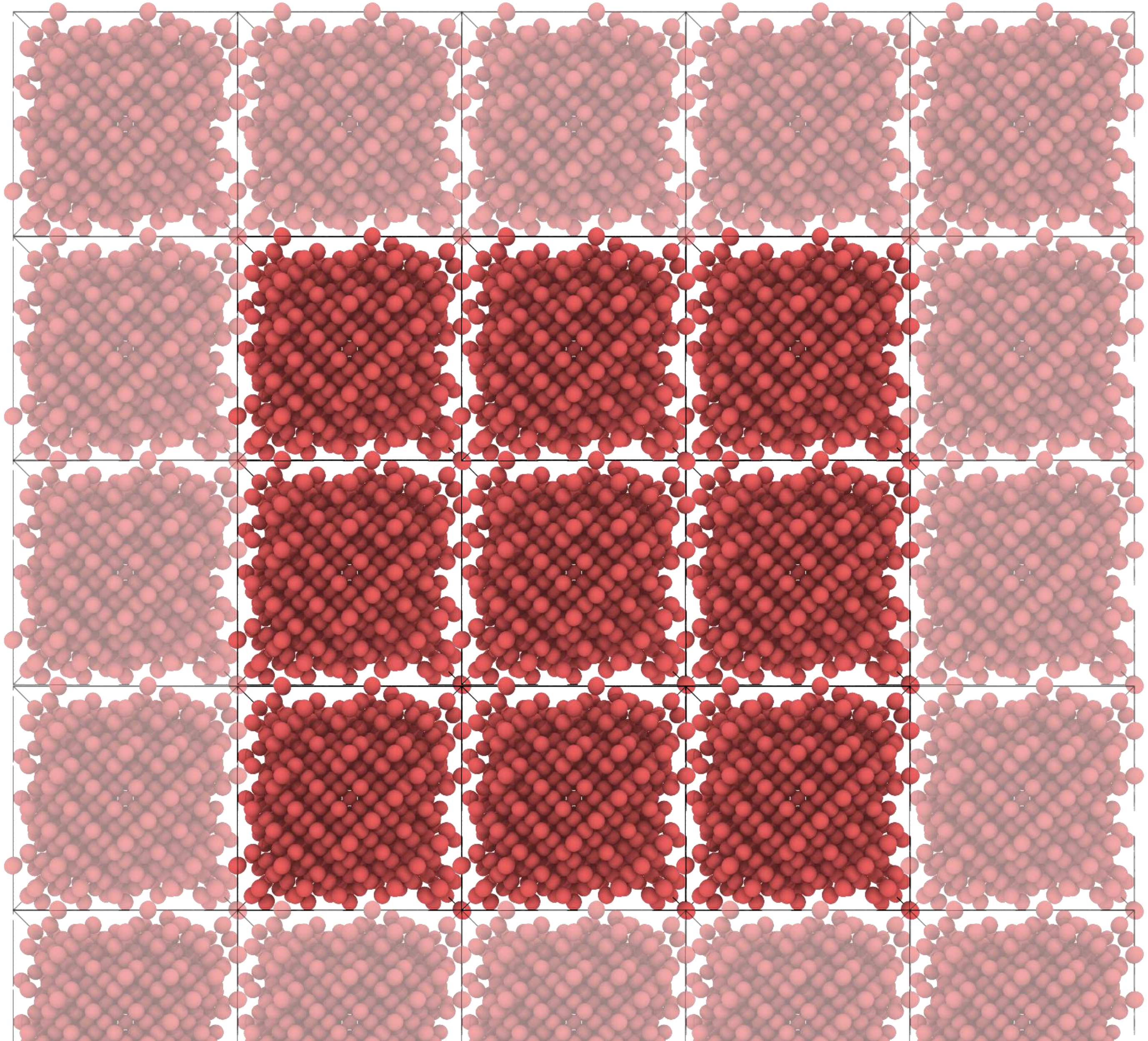
MPI Rank 2

LEVERAGING LOCALITY IN MPI COMMS

The central rank must communicate with its immediate neighbors

but not the ranks beyond

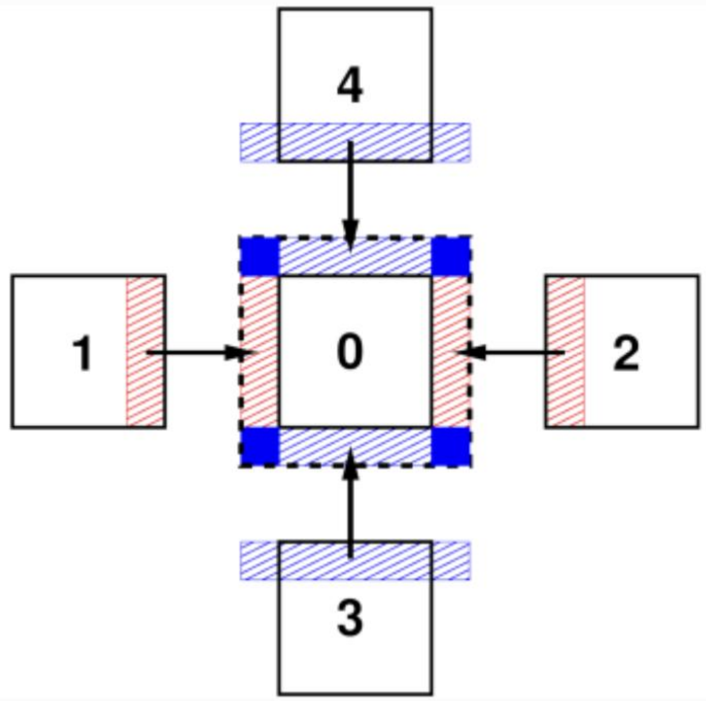
So we could use targeted messages with MPI_Sendrecv



We only have to re-assign atoms as frequently as our neighbor list updates

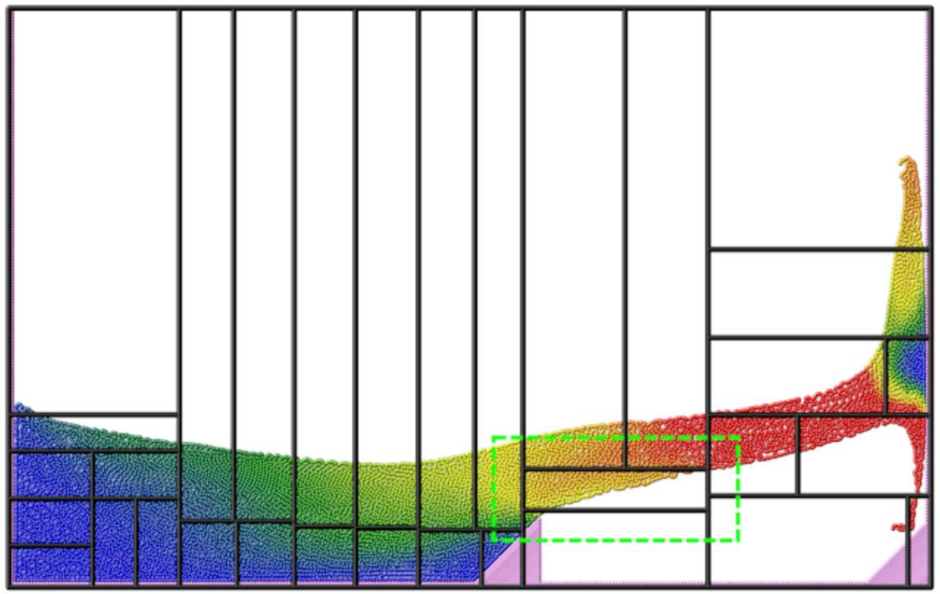
IMPROVED DOMAIN DECOMPOSITION

Trading more communication steps
for less data/ranks per step



https://docs.lammps.org/Developer_par_comm.html

Load balancing for heterogenous
system structure



https://docs.lammps.org/Developer_par_part.html

HOW FAR DOES THIS GET US?

- Exercise 3

- This is not so bad, we can begin to see some melting!

SHARED MEMORY PARALLELISM ON GPU

- Similarly to OpenMP parallelization, we can leverage the GPU for looping operations

- We just need to write some CUDA equivalent to our previous operations...

HOW FAR DOES THIS GET US?

- Exercise 4

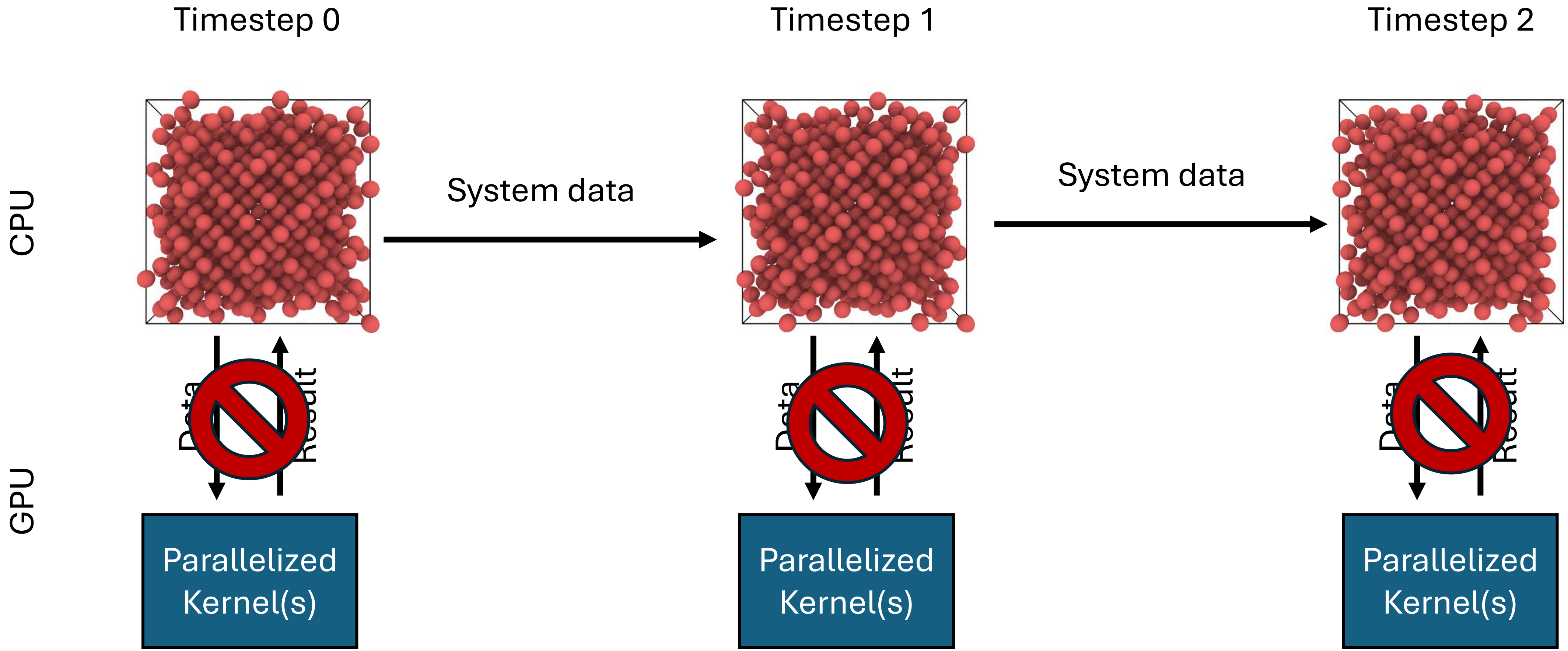
- Now, we can really get somewhere, even in 1 minute!

PUTTING IT ALL TOGETHER

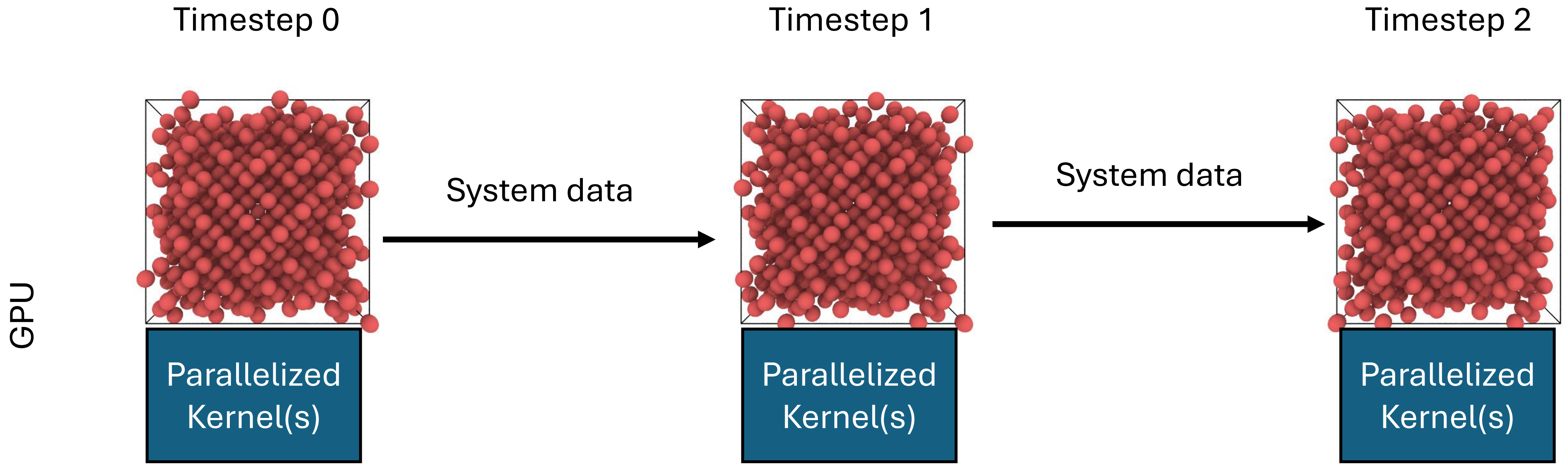
- Exercise 5

- The ice can be melted!

THE NEXT STEPS



THE NEXT STEPS



FURTHER REFERENCES

- Understanding Molecular Simulation by Frenkel & Smidt
- Molinero, V., & Moore, E. B. (2009). Water modeled as an intermediate element between carbon and silicon. *J. Phys. Chem. B*, *113*(13), 4008-4016.
- Names in Molecular Dynamics Engines
  - LAMMPS
  - Gromacs
  - NAMD
  - OpenMM
  - HOOMD-Blue
- High Performance MD
  - LAMMPS-Kokkos
  - ACM Gordon Bell Prize Winners and Finalists
- Names in Force Field Development
  - AMBER
  - OPLS
  - MARTINI
  - NequIP
  - ACE